Introduction to Labour Market Economics

ECON 300 - Labour Economics

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Winter 2026

Learning Objectives

- Understand that earnings, hours, and hourly wages are unevenly distributed.
- Learn how supply and demand determine equilibrium wages and employment.
- Identify how laws and institutions affect labour market outcomes.
- Explore different data types (time-series, cross-section, panel).
- Understand econometric tools like regression analysis.

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Main Players and Roles: Individuals

- Decide when to enter or exit the labour force.
- Choose how much education and training to pursue.
- Determine hours of work, job type, and location.
- Evaluate wages, benefits, and work-life balance.

Example: Deciding between full-time work or continuing education involves weighing opportunity costs and future wages.

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- **Legislative Environment:** Minimum wage, parental leave, safety standards.
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- Provides EI, training, pensions, and insurance.
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Labour Market Economics

Definition: Labour economics studies how wages, employment, and unemployment are determined by the interaction of labour supply and demand.

Key Idea

The labour market links people's time and skills with firms' demand for productive inputs.

Dimensions of Labour Supply

Quantity Dimensions

Population growth and participation rates.

Quality Dimensions

Education, health, and productivity.

Income Maintenance Programs

Welfare, disability, and employment insurance affect incentives to work.

Dimensions of Labour Demand

- Labour costs (wages, benefits, mandated costs).
- Demand for output (product demand drives labour demand).
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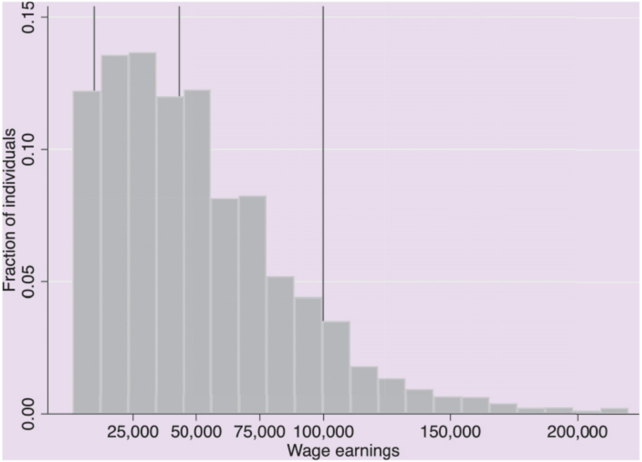
Sources of Income for Canadians, 2017

Table: Table 1.1: Sources of Income for Individual Canadians, 2017

	Full Sample	**Men**	**Women**
Earnings	41,348	50,824	31,910
Investment income	2,542	2,906	2,179
Pension income	1,262	1,484	1,040
Other income	952	971	933
Gov. transfers	3,728	2,495	4,955
Subtotal	49,831	58,681	41,017
Income taxes	8,391	11,168	5,625
After-tax total	41,440	47,512	35,392
Hours worked	1,411	1,593	1,230
Hourly wage	29.77	32.57	26.67

Figure 1.1: Distribution of Individual Labour Earnings (2017)



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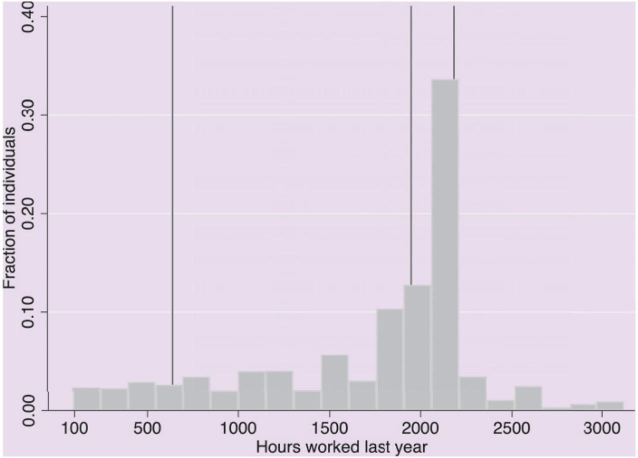

FIGURE 1.2 The Distribution of Individual Annual Hours Worked, 2017

Figure 1.3: Distribution of Average Hourly Earnings (2017)

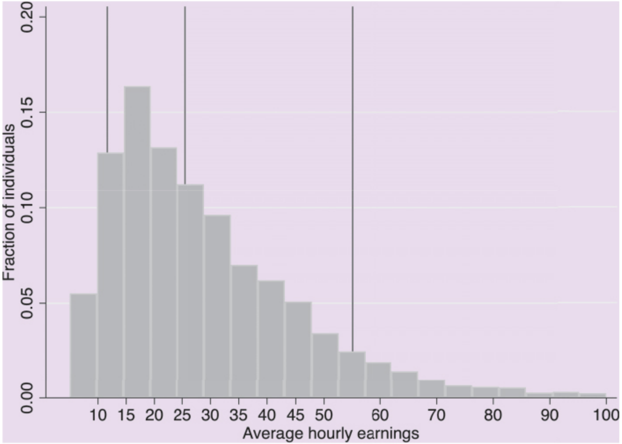

FIGURE 1.3 The Distribution of Individual Average Hourly Earnings (Wages), 2017

Neoclassical Model of Labour Supply and Demand

- Workers supply labour to maximize utility.
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Figure 1.4: Wages and Employment in a Competitive Market

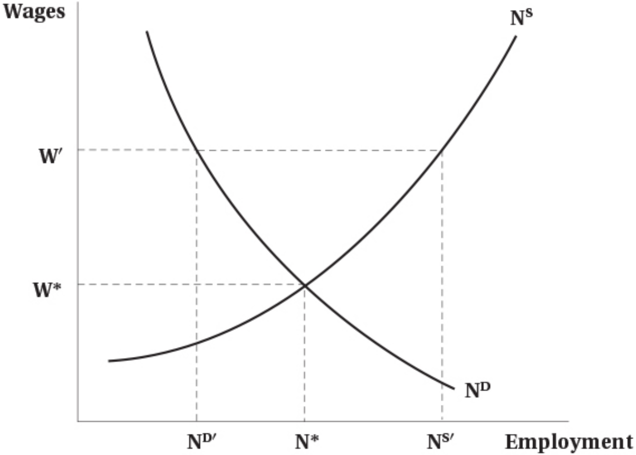
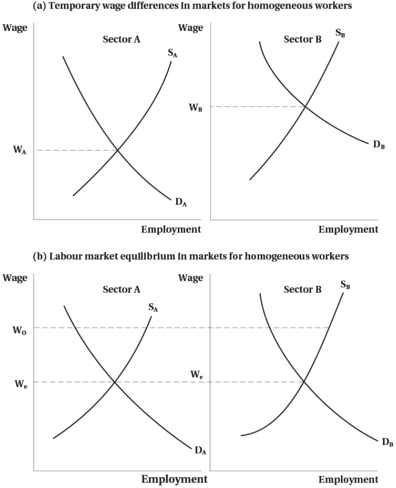


FIGURE 1.4 **Wages and Employment in a Competitive Labour Market**

Figure 1.5: Equilibrium Wages Across Two Sectors



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Current Policy Issues

- **Labour Supply:** Impact of EI, welfare, and childcare.
- **Labour Demand:** Effects of trade, outsourcing, and regulation.
- **Wage Structures:** Gender gaps, benefits, and equity.
- **Unemployment:** Youth and cyclical unemployment trends.

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Figure 1.6: COVID-19 Employment Losses Across Earning Quartiles

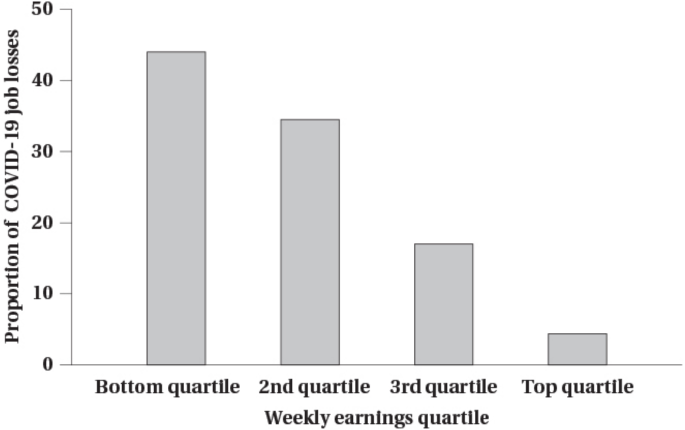

FIGURE 1.6 The Distritribution of COVID-19 Employment Losses Across Earning Quartiles

Labour Market vs. Other Markets

- Actors pursue diverse goals beyond profit.
- Influenced by legal, institutional, and social factors.
- Wages serve as price, signal, and incentive simultaneously.

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- Outcomes: wages, employment, hours.
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Appendix 1A: Regression Analysis and Research Design

- Data types: time-series, cross-section, and panel.
- Empirical tools link labour outcomes with determinants.
- Example: NHL player salaries and performance.

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Figure 1.A1: Supply and Demand – “Star” vs. “Grinder” Players

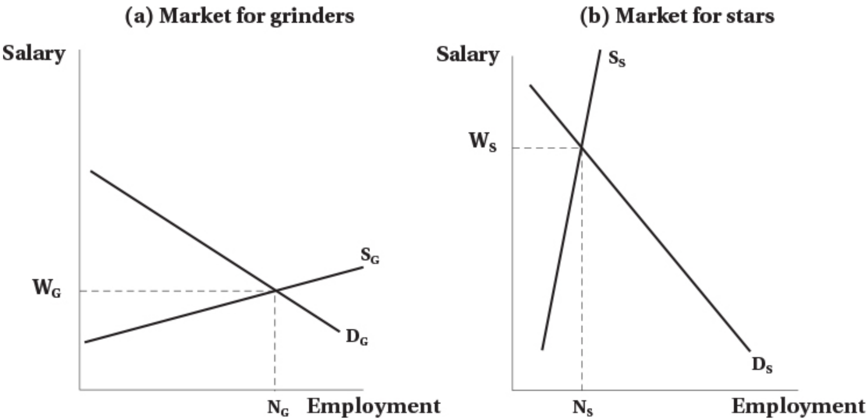

FIGURE 1.A1 Supply and Demand and the Relative Earnings of Star versus Grinder Hockey Players

Figure 1.A2: NHL Player Salaries by Goals Scored (2018–19)

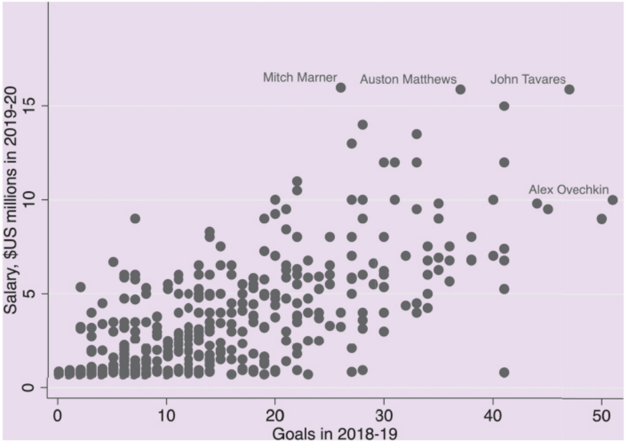

FIGURE 1.A2 NHL Player Salaries by Goals Scored in the Previous Year

Figure 1.A3: Log Salary and Goals with Regression Line (2018–19)

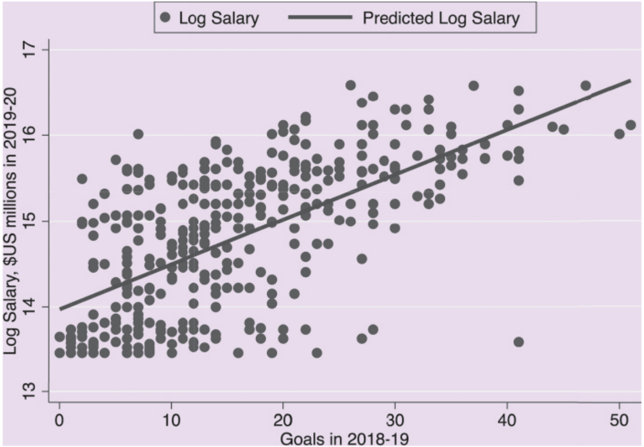

FIGURE 1.A3 Logarithm of NHL Player Salaries by Goals Scored in the Previous Year, and Fitted Regression Line

Tabulation of NHL Player Salaries by Goals Scored

Table: Table 1.A1: NHL Player Salaries by Goals Scored, 2018–19

Goals	Sample Size	Salary (millions)	$\ln$(Salary)
0–9	121	1.93	14.20
10–19	133	3.13	14.73
20–29	76	5.53	15.32
30–39	32	7.51	15.76
40–49	11	9.03	15.82
50+	2	9.50	16.07

Estimated Effects of Performance on Salary (OLS, 2019–20)

Table: Table 1.A2: OLS Estimates of Player Salary Determinants

	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	685,592	247,395	197,563	13.972	13.867	13.859
Goals	195,498	88,605	90,550	0.052	0.027	0.027
Assists		98,218	98,902		0.024	0.024
Plus/Minus			-8,156			-0.001
R^2	0.44	0.54	0.54	0.39	0.47	0.47

Appendix 1B: Research Designs in Labour Economics

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- Non-experimental: Regression, DiD, natural experiments.
- Purpose: Estimate counterfactuals—what would happen without policy change.

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